

Artificial Intelligence and Retrieval Augmented Generation (RAG)

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on building intelligent systems that can perform tasks requiring human intelligence. AI systems can learn from data, recognize patterns, understand language, and make decisions.

Machine Learning

Machine Learning is a subset of Artificial Intelligence that enables computers to learn from data. Instead of using manually written rules, machine learning algorithms identify patterns from examples and use those patterns to make predictions.

Types of Machine Learning

Supervised Learning:

Supervised learning uses labeled data where the expected output is already known. Examples include spam detection, image classification, and price prediction.

Unsupervised Learning:

Unsupervised learning works with unlabeled data and discovers hidden patterns. Examples include customer grouping and document clustering.

Reinforcement Learning:

Reinforcement learning allows an agent to learn through interaction with an environment by receiving rewards and penalties.

Deep Learning

Deep Learning uses artificial neural networks with multiple layers to learn complex patterns. It is widely used in computer vision, speech recognition, and natural language processing.

Large Language Models

Large Language Models (LLMs) are AI models trained on huge amounts of text data. They can understand and generate human-like language. Examples include GPT, Gemini, Claude, and Llama.

Retrieval Augmented Generation

Retrieval Augmented Generation (RAG) is a technique that improves Large Language Models by connecting them with external knowledge sources. Instead of depending only on training data, RAG retrieves relevant documents and provides them as context to the AI model.

RAG Pipeline Steps

1. Document Loading:

Documents such as text files, PDFs, websites, and databases are loaded into the application.

2. Text Splitting:

Large documents are divided into smaller chunks to improve retrieval performance.

3. Embeddings:

Text chunks are converted into numerical vectors that represent their meaning.

4. Vector Database:

Embeddings are stored in vector databases such as FAISS and ChromaDB for similarity search.

5. Retrieval:

When a user asks a question, the system finds the most relevant document chunks.

6. Generation:

The retrieved information is provided to the language model, which generates the final answer.

LangChain

LangChain is a framework used to build applications powered by Large Language Models. It provides components for document loaders, text splitters, embeddings, retrievers, and chains.

PDF Document Loaders

PDF document loaders allow RAG systems to extract text from PDF files. LangChain provides tools such as PyPDFLoader to load PDF documents and convert them into document objects.

Applications of RAG

RAG is used in customer support chatbots, educational assistants, healthcare information systems, enterprise search systems, and document question-answering applications.

Conclusion

RAG combines document retrieval with powerful language models to create accurate and reliable AI systems. It reduces hallucination and allows AI applications to use private and updated information.